

Math 2710 Schedule

Schedule

Date

Sections

Topics

Tu 09/01

§1.1

Sets, elements, and set notation; set builder notation

Th 09/03

§1.2–1.3

Cartesian products; subsets; counting subsets

Tu 09/08

§1.4–1.5

Power sets; union, intersection, and difference

Th 09/10

§1.6–1.8

Complement; Venn diagrams; indexed sets

Tu 09/15

§2.1–2.3

Statements and truth values; connectives; conditional statements

Th 09/17

§2.4–2.6

Biconditionals; truth tables; logical equivalence

Tu 09/22

§2.7–2.8

Quantifiers; paired quantifiers; implications with open sentences

Th 09/24

§2.9–2.11

Translating into logic; negations; deduction rules

Tu 09/29

Review

Th 10/01

First Midterm

Tu 10/06

§3.1–3.2

Multiplication principle; addition principle

Th 10/08

§3.3–3.5

Permutations; combinations; Pascal's triangle and binomial theorem

Tu 10/13

§4.1–4.2

Theorems and definitions; division algorithm

Th 10/15

§4.3

Direct proofs; examples

Tu 10/20

§4.3–4.4

Using definitions in proofs; more direct proofs; proof by cases

Th 10/22

§5.1–5.2

Proof by contrapositive; congruences; congruence arithmetic

Tu 10/27

§6.1, §7.1

Proof by contradiction; Euclid's proof of infinitely many primes; if-and-only-if proofs

Th 10/29

§7.3–7.4

Existence proofs; Euclid's algorithm; uniqueness proofs; constructive and non-constructive proofs

Tu 11/03

Review

Th 11/05

Second Midterm

Tu 11/10

§8.1–8.3

Proving set membership; set inclusion proofs; set equality proofs

Th 11/12

§9.1–9.3

Counterexamples; disproving universal statements; disproving existence claims

Tu 11/17

§10.1

Introduction to induction; induction proofs; more examples

Th 11/19

§10.2–10.5

Strong induction; strong induction continued; fundamental theorem of arithmetic; Fibonacci numbers; well-ordering principle

Tu 11/24

Thanksgiving Break

Th 11/26

Thanksgiving Break

Tu 12/01

§11.1–11.2

Relations; reflexive, symmetric, and transitive properties

Th 12/03

§11.3–11.6

Equivalence relations; partitions; integers mod n ; relations between sets

Tu 12/08

§12.1–12.2

Functions; injective, surjective, and bijective functions

Th 12/10

§12.3–12.6

Pigeonhole principle; composition; inverse functions; image and preimage